

Optimization 5 - Principal Component Analysis

Agenda

- Recap
- Variants of gradient descent
 - * Vanilla GD
 - * SGD
 - * Batch GD
- Dimensionality reduction
- Input Output relationship
- Example height vs age
- Visual understanding of PCA
- Implementing PCA
- Mathematical approach of optimizing PCA
- What is eigen vector.

Recap

5 lectures

5 lectures

linear algebra

line — $y = mx + c$

$$w_0 x + w_1 y + w_0 = 0$$

norm = magnitude = length

$$\hookrightarrow \sqrt{w_0^2 + w_1^2}$$

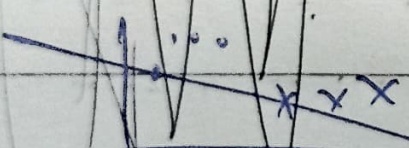
dot product $\rightarrow w^T x$

$$\text{unit vector} = w / \|w\| = \hat{w}$$

projectⁿ of vectors

Perceptron algorithm

start with random line



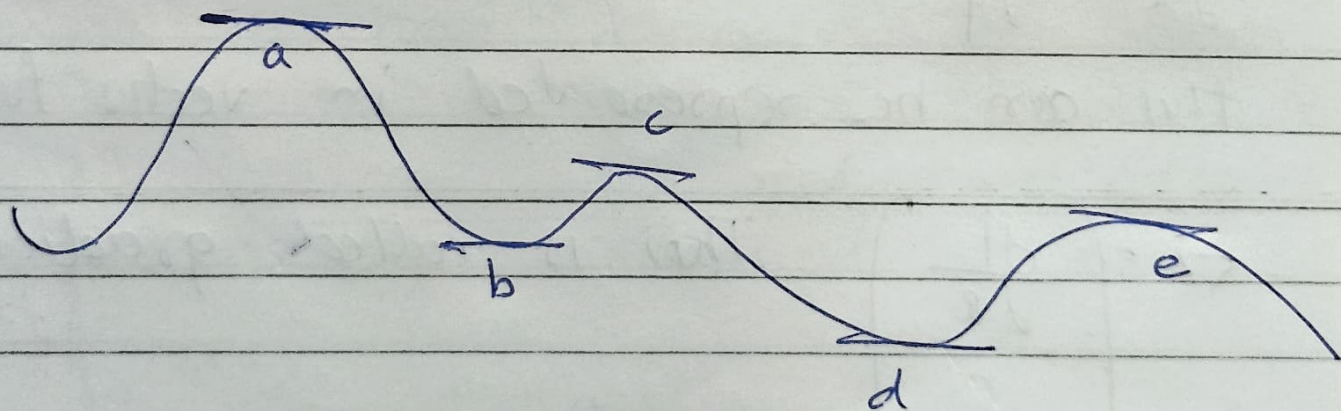
find mis-classifiedⁿ

$$d = \max \sum \frac{w^T x + w_0}{\|w\|} y_i$$

Optimization

Now we have a fnet^m (loss fnet^m) we need to minimize it, in other words we need $\bar{w}$ & w_0 where loss fnet^m is minimum, this is called *exymun* here loss fnet^m is fnet^m as other fnet^m .

The minimum/maximum of a fnet^m can be found by slope/derivative. As shown in fig. below



Here a, b, c, d, e is slope and a, c, e are local ~~minima~~ maxima & b, d are local minima.

so if we do $\frac{df}{dx} = 0$ then we will get

x , & I think it is argmin.

if function have more than 1 variable then we need partial derivative

$$\frac{df}{dx}, \frac{df}{dy} \dots$$

this can be represented in vector form

$$\nabla = \begin{pmatrix} \frac{df}{dx} \\ \frac{df}{dy} \\ \vdots \end{pmatrix} \quad \text{This is called gradient}$$

If the directⁿ of gradient is upward then it is called "gradient ascent".

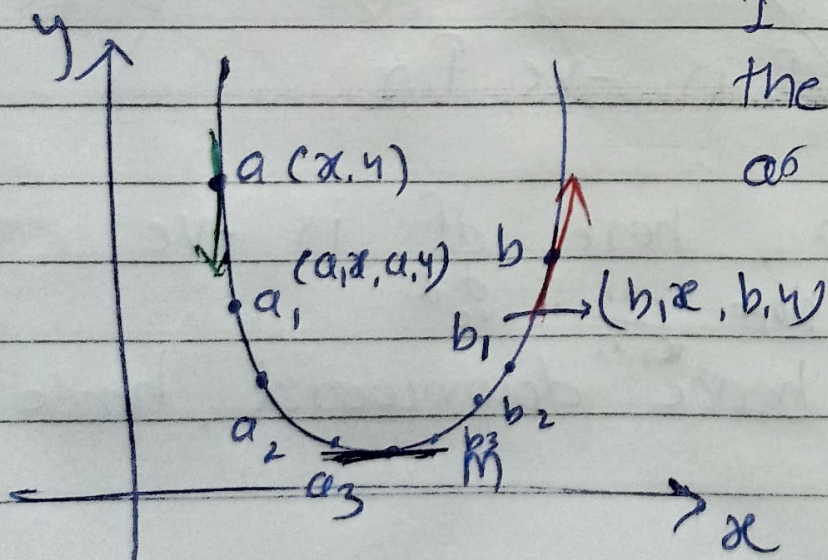
If the directⁿ of gradient is downward then it is called "gradient descent".

As we need to find minima, so common term & used is "gradient descent".

More on "gradient ascent" & "gradient descent".

Suppose we have a curve as below.

I want to calculate the gradient (slope) at point a & b.



$$\text{gradient} = \nabla = \begin{bmatrix} \frac{df}{dx} \\ \frac{df}{dy} \end{bmatrix}$$

$$\frac{df}{dx} = \lim_{\Delta \rightarrow 0} \frac{f(x+\Delta, y) - f(x, y)}{\Delta}$$

$$\frac{df}{dy} = \lim_{\Delta \rightarrow 0} \frac{f(x, y+\Delta) - f(x, y)}{\Delta}$$

~~From the figure we can say if Δ is +ve then the f is +ve for in y direction at both a & b points~~

~~if Δ is +ve then f is -ve for~~

consider point a, here $\frac{df}{dx}$ is -ve

hence slope $\frac{y_2 - y_1}{x_2 - x_1}$ is -ve

& $\frac{df}{dy}$ is +ve, $\uparrow$ hence downward, hence

gradient decr

consider point b, here $\frac{df}{dx}$ is +ve
& $\frac{df}{dy}$ is +ve

hence upward, hence gradient ascent.

hence slope $\frac{y_2 - y_1}{x_2 - x_1}$ is +ve

+ve slope $\Rightarrow$  1st & 3rd quadrant

-ve slope $\Rightarrow$  2nd & 4th quadrant

~~Here we need to find w where slope is 0, the eqn at w~~

$$\text{w}_{\text{new}} = \text{w}_{\text{old}} - \eta \frac{df}{dw}$$

~~This is called gradient descent rule~~

Here our goal is to find the point m where the function is minimum.

case I

can

If we select point ~~a~~ a or b & try to reach to point m .

→ Suppose we select the point a . Then we calculate $\frac{df}{dx}$ & $\frac{df}{dy}$ at that point.

here $\frac{df}{dx}$ is -ve & $\frac{df}{dy}$ is +ve, so

-ve slope. we ~~can~~ consider ~~get~~ new point

$$a_1 = a - \eta \frac{df}{dw}$$

$$a_{1,x} = a_x - \eta \frac{df}{dx}$$

$$a_{1,y} = a_y - \eta \frac{df}{dy}$$

$$\eta = 0.001$$

here a, x & a, y will give us
a point

here a, x will be greater than a, x
as $\frac{df}{dx}$ is -ve

here a, y will be less than y
as $\frac{df}{dy}$ is +ve

so a will be on the right of a
on the curve as shown in figure

Again we get new point

$$a_2 = a_1 - \eta \frac{df}{dw}$$

$$a_2 x = a_1 x - \eta \frac{df}{dx}$$

$$a_2 y = a_1 y - \eta \frac{df}{dy}$$

again a_2 will be on sight of a ,
similar way we get a_3 & finally
minimum in point.

Note if n is large we can get point
near b , that's why need to choose
 n carefully.

case II

Suppose we select point b , we
calculate $\frac{df}{dx}$ & $\frac{df}{dy}$ at b

$\frac{df}{dx}$ +ve & $\frac{df}{dy}$ +ve, so +ve slope

there new point will be

$$b_1 = b - n \frac{df}{dy}$$

$$b_1 x = b_x - n \frac{df}{dx}$$

$$b, y = b_y - n \frac{dF}{dy}$$

b_x & b_y will give point point
 b_x
here b_x will be smaller than x
 b_y " " " " " "

as both $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$ are +ve

so b_1 will be on left of b
as shown - in graph

Again similarly we get b_2, b_3
which are on left of b
& finally m .

So $\omega_{\text{new}} = \omega_{\text{old}} - n \frac{dF}{d\omega}$

is called "gradient descent" because by this formula we get the w which descends the gradient to 0.

loss function to minimize

$$= \min \left(- \sum \frac{\omega^T x + \omega_0 y_i}{\|\omega\|} \right)$$

Need to get rid of $\|\omega\|$. If we put $\|\omega\| - 1 = 0$ constraint, then it is a constrained optimization problem & can be solved by Lagrangian multiplier

$$= \min \left(- \sum \frac{\omega^T x + \omega_0 y_i}{\|\omega\|} - \lambda (\|\omega\| - 1) \right)$$

Need to understand more on Lagrangian multiplier in depth.

→ Need to understand how this formula came which consider long term multiplier

Vanilla gradient descent $\frac{dL}{d\omega}$

$$\bar{\omega}^{(t+1)} = \bar{\omega}^{(t)} - \eta \left(- \sum_{i=1}^n y_i x_i + \lambda \frac{\bar{\omega}}{\|\bar{\omega}\|} \right)$$

$dL/d\omega$

$$\omega_0^{(t+1)} = \omega_0^{(t)} - \eta \left(- \sum_{i=1}^n y_i \right)$$

if n is big then too much expensive. $\frac{dL}{d\omega} \approx O(n)$.

Now consider only single point instead of whole data set

$$\bar{w}^{(t+1)} = \bar{w}^{(t)} - \eta \left(y_i x_i + \lambda \frac{\bar{w}}{||\bar{w}||} \right)$$

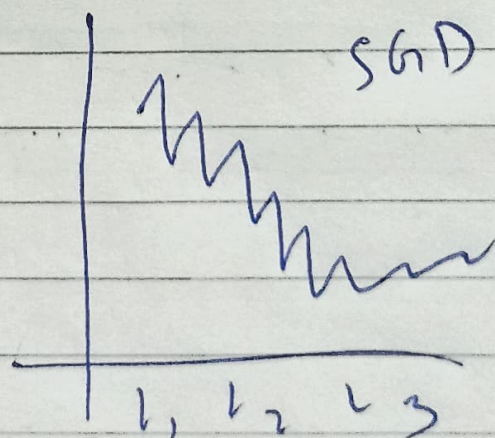
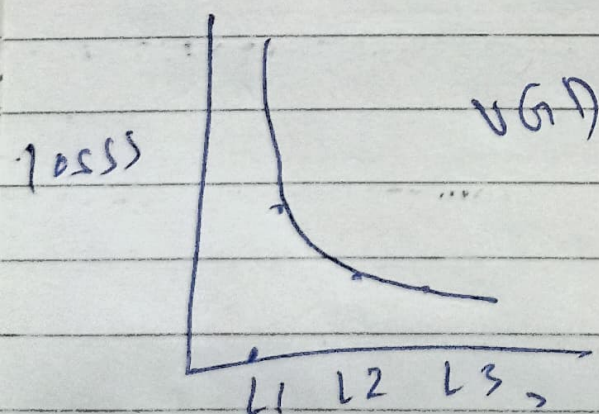
$$w_0^{(t+1)} = w_0^{(t)} - \eta (-y_i)$$

$\uparrow$
 $o(1)$

Thus is called
Stochastic Gradient Descent

which means instead of taking all points take points one by one & update $\bar{w}$ & w_0

disadvantage: -



loss will fluctuate a lot

Stochastic meaning $\rightarrow$ random, involving random variable. Here in SGD ~~we are taking~~ instead of taking whole data set we are taking a single random point in data set hence SGD.

GD & SGD are 2 extreme method something in betⁿ will be good. Such as taking sample batch of data within whole data set.

which is called "mini batch GD".

$$\bar{w}^{(t+1)} = \bar{w}^{(t)} - \eta \left(\sum_{i=1}^K y_i x_i + \lambda \frac{\bar{w}}{\|\bar{w}\|} \right)$$

$$w_0^{(t+1)} = w_0^{(t)} - \eta \left(- \sum_{i=1}^K y_i \right)$$

$K=1$ SGD

$1 \leq K < n$ minibatch GD

$K=n$ GD

if $n = 1000$

$K = 10$

batches = 10

updating w, w_0 will be 10 times.

So we can update w, w_0 by
VGP (considering all points)

↳ Here weight will be updated after
considering all points

We can do this multiple times

it is called epoch

SGD $\rightarrow$ here weight, w_0 are updated after considering random point. We can do this n times to cover all point.

We can do above multiple times it is called epoch.

minibatch SGD

here weight, w_0 are updated after consider a sample data set of k points. We can do this n/k times to cover all point.

We can do above multiple times it is called epoch.

PCA $\rightarrow$ reduce dimension.

$$x_i \in \mathbb{R}^d \rightarrow \boxed{\text{PCA}} \rightarrow x_i \in \mathbb{R}^{d'}$$
$$d' < d$$

PCA

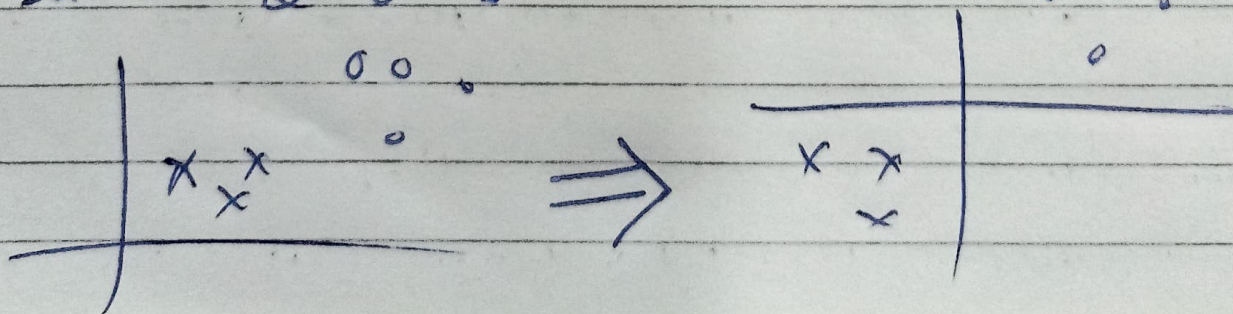
① Normalize

to bring all features in same range

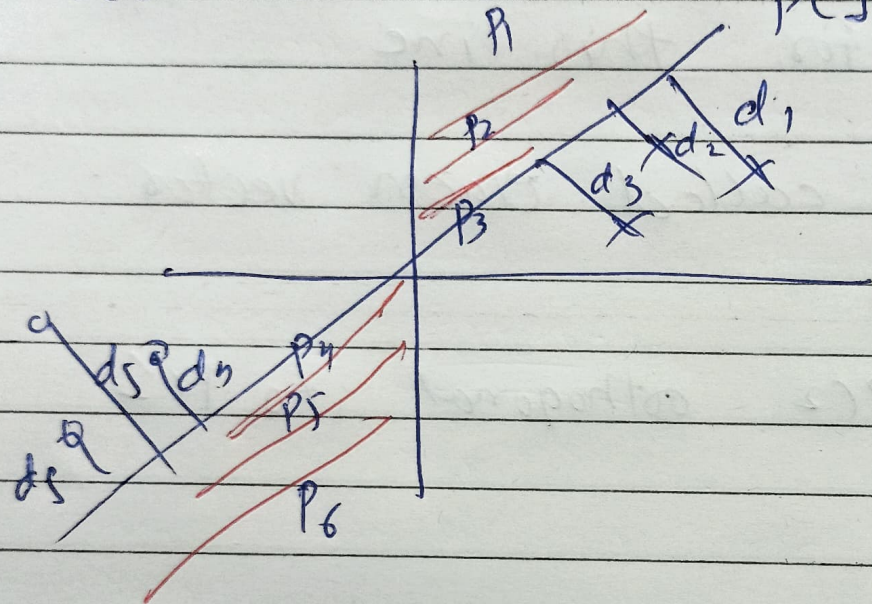
Standard normalized

$$\frac{x - \mu}{\sigma}$$

This will lead to data having
 $\mu = 0$ & $\sigma = 1$



② best fit line



here $d_1 + d_2 + d_3$ should be max
 $P_1 + P_2 + P_3$ " " min

III

③* once got best fit line, calculate unit vector for this line

unit vector is called eigen vector.

IV

④* calculate PC2 orthogonal to PC1